

Bending Light: Investigating Refraction

SimuPhysics Refraction Simulation · SimuPhysics

Years 8-9

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://simuphysics.com/view/light-refraction-simulation.html>

Curriculum links: AC9S8U05 · AC9S9U04 · waves

Learning intentions

- I can describe what happens to a light ray when it crosses from one medium into another.
- I can measure angles of incidence and refraction and compare them across different materials.
- I can explain refraction using a wave model of light.

Getting started

Open the SimuPhysics Refraction Simulation. Start with light travelling from air into water. Use the angle control (or drag the incoming ray) to change the angle of incidence, and use the protractor or angle readout to measure the angle of refraction. Keep the two angle values visible as you work through the tasks.

Tasks

1. Set the angle of incidence to a small angle (close to the normal line). Record the angle of incidence and the resulting angle of refraction.

2. Increase the angle of incidence to a larger value. Record the new angle of incidence and angle of refraction.

3. Compare your two results. Does the light ray bend towards the normal line or away from it when it enters the denser medium (e.g. water)? Explain what this tells you about the speed of light in each medium.

4. Test five different angles of incidence going from air into your chosen medium. Record each pair of angles.

Angle of incidence (°)	Angle of refraction (°)

5. Change the second medium to a different material (e.g. glass instead of water, if available). Using the same angle of incidence as one of your earlier trials, record the new angle of refraction. Is the bending greater or smaller than before?

6. Using the wave model of light, explain why light bends when it slows down entering a denser medium. You may describe what happens to the wavefronts as they cross the boundary at an angle.

7. Using two of your recorded angle pairs, calculate $\sin(\text{angle of incidence}) \div \sin(\text{angle of refraction})$ for each. What do you notice about these two ratios?

Working:

8. Predict what would happen to the angle of refraction if the light ray hit the boundary exactly along the normal line (0° angle of incidence). Test your prediction in the simulation.

Extension (optional)

Explain why a straw in a glass of water appears to 'bend' or 'break' at the water's surface when viewed from the side, using what you have learned about refraction in this simulation.
